

UNIVERSITY OF DHAKA

**PGBM: A Proactive Golden-Hour Bridge Model
for Time-Critical Post-Earthquake Survivor
Assistance with Joint Task Allocation, 3D
Routing and Dynamic Recourse**

by

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“In the middle of difficulty lies opportunity.”

— Albert Einstein

Abstract

Post-earthquake search-and-rescue operations are highly time-sensitive, and UAVs can support emergency response by rapidly surveying hazardous areas and locating survivors. However, locating a survivor does not guarantee immediate assistance, as ground SAR teams may be delayed by blocked roads, damaged infrastructure, unstable debris, and limited responder availability. These challenges are further complicated by the limited payload, battery capacity, and mission time of UAVs, as well as the need for safe and feasible 3D routes. Existing studies have addressed UAV-based survivor detection, reconnaissance, delivery, and route planning, but these capabilities are often considered separately within the disaster-response process. This thesis proposes the **Proactive Golden-Hour Bridge Model (PGBM)**, which introduces dedicated responder UAVs to deliver lightweight emergency supplies during the period between survivor detection and ground rescue. PGBM jointly considers task selection, responder assignment, service order, and feasible 3D routing while accounting for resource and energy constraints. It also uses a time-dependent service value to represent the effect of delayed assistance and includes a controlled task-replacement mechanism for newly arriving survivor-assistance tasks during active missions.

Keywords: Proactive Golden-Hour Bridge Model (PGBM), UAV-assisted disaster response, responder UAVs, survivor-assistance tasks, task allocation, 3D routing, dynamic task-replacement recourse, resource-aware optimization, post-earthquake response.

Contents

Declaration Form	i
Abstract	iii
Acknowledgements	iv
List of Figures	vii
List of Tables	viii
1 Introduction	1
1.1 Background	1
1.2 Motivation	2
1.3 Problem Statement	4
1.4 Objectives	6
1.5 Research Contributions	6
1.6 Research Challenges	7
1.7 Organization of the Thesis	8
2 Related Work	10
2.1 UAV-Based Disaster Perception and Search	10
2.2 UAV Delivery, Resource-Aware and Adaptive Mission Planning	12
2.3 Comparative Analysis and Research Gap	13
2.4 Summary	14
3 Proposed Methodology	15
3.1 Main Notation	17
3.2 Decision Variables	19
3.3 Time-Dependent Service Value	19
3.4 Optimization Problem	20
3.5 Joint Task Allocation and Three-Dimensional Routing	21
3.6 Travel and Task Completion Time	22
3.7 Relief Capacity and Payload Evolution	23
3.8 Energy and Battery Feasibility	24
3.9 Task-Replacement Recourse	25

4	Implementation	28
4.1	Implementation Overview	28
4.2	Data Sources and Scenario Construction	29
4.3	Synthetic Survivor-Assistance Task Generation	29
4.4	Three-Dimensional Operating Environment	30
4.5	Responder-UAV and Payload Configuration	31
4.6	Optimization and Recourse Workflow	31
4.7	Planned Evaluation Metrics	32
5	Experimental Results	33
5.1	Temporal Service-Value Decay Experiment	33
5.2	Results Analysis	38
5.3	Summary of the Experimental Results	38
6	Conclusions	39
6.1	Research Summary	39
6.2	Future Work Plan	40
	Bibliography	41

List of Figures

1.1	Baseline passive reconnaissance scenario in which scout UAVs report survivor locations to a command center, after which assistance depends on the availability and accessibility of ground SAR teams. . .	2
1.2	Proposed golden-hour bridge scenario in which responder UAVs provide temporary relief delivery between survivor detection and physical rescue by ground SAR teams.	3
3.1	Overall PGBM system architecture showing scout-UAV detection, command-center task and UAV-state management, online responder assignment, energy-aware route planning, and mission-state feedback.	16
5.1	Reference policy curves used to guide service-value calibration. . . .	34
5.2	Fitted service-value curves for the four candidate functions.	35
5.3	Fitting error comparison for the candidate service-value functions. .	36
5.4	Fitted common exponential service-value decay curves. Markers indicate the reference policy values used for calibration.	37

List of Tables

2.1	Comparison of selected UAV disaster-response studies with PGBM.	13
3.1	Main notation used in the PGBM formulation.	17
4.1	Planned implementation data components.	32
5.1	Reference policy values used for service-value calibration.	34
5.2	Candidate service-value functions evaluated in the experiment. . . .	35
5.3	Comparison of candidate service-value decay functions.	36

Acronym	Meaning
3D	Three-Dimensional
5G	Fifth-Generation Mobile Network
AI	Artificial Intelligence
AoA	Angle of Arrival
BIC	Bayesian Information Criterion
BLE	Bluetooth Low Energy
GMM	Gaussian Mixture Model
GPS	Global Positioning System
LiDAR	Light Detection and Ranging
LoS	Line of Sight
NLoS	Non-Line of Sight
PGBM	Proactive Golden-Hour Bridge Model
PSO	Particle Swarm Optimization
RGB	Red, Green, and Blue
RSSI	Received Signal Strength Indicator
SAR	Search and Rescue
SLAM	Simultaneous Localization and Mapping
ToF	Time of Flight
UAV	Unmanned Aerial Vehicle

Symbol	Description
$\mathcal{I}(t), \mathcal{U}(t), \mathcal{R}$	Eligible tasks, available responder UAVs, and relief-item types at planning time t
i, u, r, m, n	Task, UAV, relief-item type, and waypoint or flight-link indices
$\mathcal{N}, \mathcal{E}$	Feasible 3D UAV waypoints and directed flight links in the operating space
$\mathbf{p}_i, \mathbf{p}_m$	Detected task position and 3D coordinate of waypoint m
m_i	UAV-accessible drop-off waypoint associated with task i
b_u^s, b_u^r	Logical start and return base nodes of UAV u at the same physical base
s_i, t_i, δ_i	Operational severity score, detection time, and service duration of task i
$\lambda, v_i(\tau)$	Common temporal decay rate and time-dependent service value of task i at time τ
x_{iu}, y_{mnu}	Binary task-assignment and flight-link selection decision variables
τ_{mu}, C_i	Arrival time of UAV u at waypoint m and completion time of task i
d_{mn}, t_{mnu}	3D distance and UAV-specific travel time on flight link (m, n)
D_u, T_u	Total selected travel distance and travel time of UAV u
$q_{ir}(t), w_r$	Required quantity of relief item r for task i and weight of one item of type r
K_u, W_u	Parcel-count capacity and payload-mass capacity of UAV u
$M_i(t), M_u^0$	Total relief mass for task i and initial payload mass assigned to UAV u
L_{mnu}	Payload mass carried by UAV u while traversing flight link (m, n)
ν_u	Cruise speed of responder UAV u

Symbol	Description
$E_u(t), E_u^{\text{res}}$	Available battery energy and required reserve energy of UAV u
a_{mn}	Positive altitude gain along flight link (m, n)
$\alpha_u, \beta_u, \gamma_u$	Baseline, payload-dependent, and positive-ascent energy coefficients of UAV u
$P_u^{\text{srv}}, E_u^{\text{req}}$	Service or hovering power and required mission energy of UAV u
t_r, n	Recourse decision time and newly arrived task considered for replacement
$\mathcal{T}_u^{\text{rem}}(t_r), I_{ur}(t_r)$	Uncompleted tasks of UAV u and onboard quantity of item type r at recourse time
$J_{ui}^{\text{old}}, J_{ui}^{\text{new}}$	Remaining mission value before and after replacing task i with the new task
$\Delta J_{ui}, E_{ui}^{\text{new}}$	Service-value gain and required energy after the replacement action
$\mathcal{S}(t_r), \mathcal{Q}(t_r)$	Feasible candidate replacements and waiting queue of currently unserved tasks

Chapter 1

Introduction

1.1 Background

After a major earthquake, the first few hours of response are often shaped by limited access rather than by lack of need. Survivors may be trapped inside damaged buildings, under rubble, or in areas surrounded by unstable structures. At the same time, ground search-and-rescue (SAR) teams may have to work through blocked roads, collapsed access points, debris, damaged communication infrastructure, and uncertain safety conditions. Physical rescue therefore remains a ground-team responsibility, but reaching every detected survivor immediately is not always possible.

Emergency operations in such conditions are usually coordinated through a command center that must distribute limited SAR, medical, logistics, and communication resources across the affected region. The command center does not only need to know where survivors may be located; it also needs to decide which reported locations should receive immediate support when responders, supplies, and access routes are limited.

Unmanned aerial vehicles (UAVs) are useful in this setting because they can enter hazardous or partially inaccessible areas faster than ground vehicles and can provide aerial observations to a command center. Recent work has studied UAV-based disaster reconnaissance, survivor detection, localization, aerial imagery analysis, and

post-earthquake scene understanding [4, 15, 19]. In the classical workflow, the UAV mainly acts as a reconnaissance platform: it scans the affected area, reports possible survivor locations, and helps the command center build situational awareness.

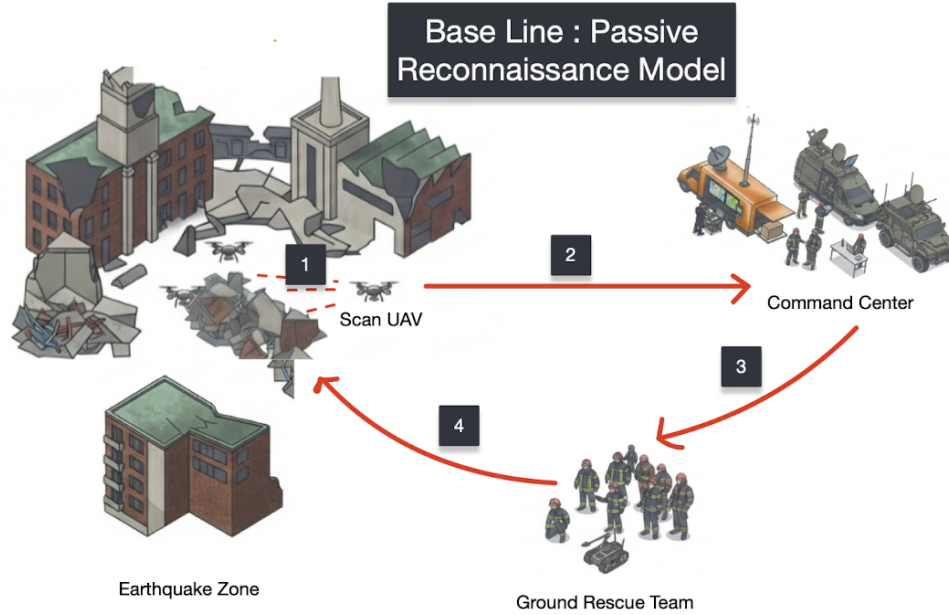


FIGURE 1.1: Baseline passive reconnaissance scenario in which scout UAVs report survivor locations to a command center, after which assistance depends on the availability and accessibility of ground SAR teams.

Figure 1.1 illustrates this baseline passive reconnaissance process. A scout UAV surveys the earthquake-affected zone and sends observations to the command center. The observations may include the suspected survivor location, aerial imagery, and contextual information about the surrounding damage. After the report is accepted, the command center depends mainly on ground SAR teams to reach the survivor physically. This workflow is valuable because it improves situational awareness, but it leaves a waiting period between detection and rescue.

1.2 Motivation

The key motivation of this thesis comes from the limitation of the passive workflow. A scan UAV can help identify where a survivor may be located, but it does not by itself deliver assistance. If ground SAR teams are delayed, the system may know

about a survivor while still being unable to provide immediate support. This is the operational gap addressed in this work.

The proposed response process therefore separates UAV operations into two roles. Scout UAVs remain responsible for reconnaissance, sensing, and reporting. Responder UAVs are introduced as a dedicated delivery layer that can carry lightweight emergency supplies such as water, food, first-aid materials, or oxygen-support items to feasible drop-off locations near detected survivors. This separation is important because reconnaissance and delivery create different operational requirements. Scout UAVs should preserve flight endurance, coverage, and sensing continuity, while responder UAVs can be loaded and routed specifically for relief delivery.

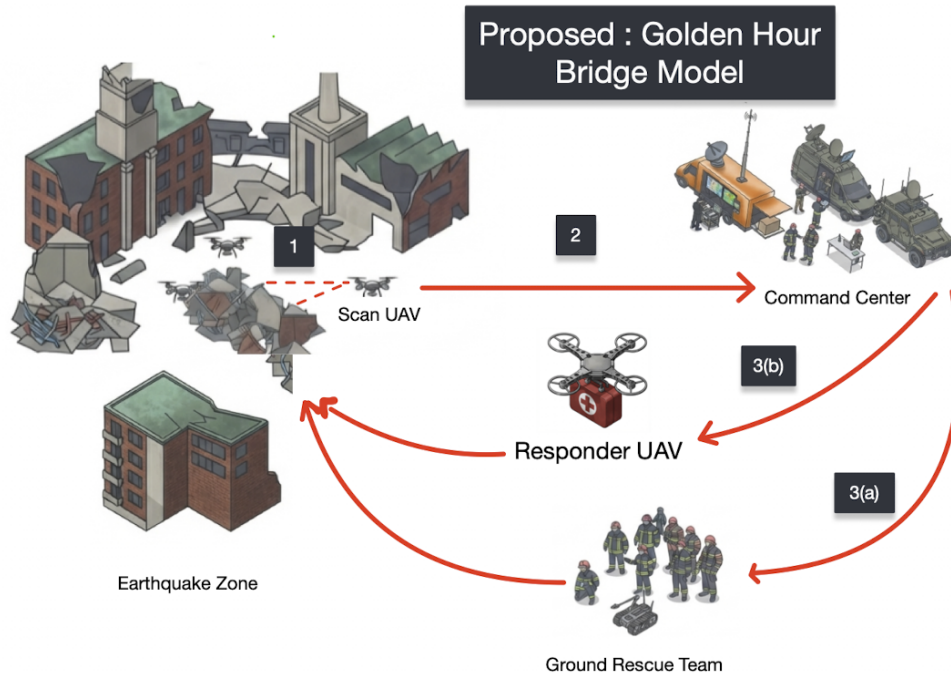


FIGURE 1.2: Proposed golden-hour bridge scenario in which responder UAVs provide temporary relief delivery between survivor detection and physical rescue by ground SAR teams.

Figure 1.2 shows the proposed golden-hour bridge scenario. Compared with the passive reconnaissance baseline, the command center now has an additional response option. After a survivor-assistance task is accepted, a responder UAV may be dispatched with suitable relief items. The responder UAV does not replace the ground SAR team and does not perform physical rescue. It provides temporary

assistance during the critical period between detection and the arrival of ground responders.

The difficulty is that responder UAV support cannot be planned greedily. A UAV has limited battery energy, flight time, payload capacity, and onboard relief items. It must also fly through a feasible three-dimensional route rather than through buildings, rubble, or no-fly regions. If several survivor-assistance tasks are waiting at the same time, choosing one task affects whether another task can still be served within the available resources. In addition, new survivor reports may arrive while a responder UAV is already in flight. These conditions motivate a planning model that treats task selection, UAV assignment, service order, routing, payload use, and energy feasibility as connected decisions rather than as separate steps.

1.3 Problem Statement

This thesis studies the responder-planning stage that begins after the command center accepts scout-UAV observations as survivor-assistance tasks. Each task represents a detected survivor or survivor group requiring a specified set of lightweight relief items. The exact physical rescue operation is still handled by ground SAR teams, and the survivor-detection algorithm itself is outside the scope of this thesis. The focus here is the decision layer that determines how available responder UAVs should deliver temporary assistance before ground rescue arrives.

The task-generation process is treated as a command-center decision. A scout UAV first reports a possible survivor location with supporting observations. The command center then validates the report, estimates the operational severity of the case, identifies the required relief items, and records the corresponding quantities. Only after this process is completed is the report treated as an eligible survivor-assistance task for responder-UAV planning.

Given a set of survivor-assistance tasks, available responder UAVs, relief-item requirements, and a feasible 3D flight network, the planning problem is to determine:

- which survivor-assistance tasks should be served;
- which responder UAV should serve each selected task;
- the order in which the selected tasks should be completed; and
- the feasible 3D route followed by each responder UAV.

The resulting missions must satisfy:

- relief-item requirements;
- parcel-count and payload-capacity constraints;
- service and travel-time requirements;
- battery and reserve-energy constraints; and
- safe return to the responder base.

Each task is associated with an operational severity and a time-dependent service value that decreases as assistance is delayed. Therefore, the planning objective is not simply to minimize flight distance or to serve the nearest task first. The objective is to maximize the total time-dependent service value obtained by the responder fleet while respecting the physical and resource limits of each UAV.

The problem also considers survivor-assistance tasks that arrive after responder missions have already started. Instead of replanning every active mission from the beginning, PGBM uses a controlled one-for-one task-replacement rule. A newly arrived task may replace one uncompleted task in an active mission only when the replacement is feasible and improves the remaining mission value. Actions that have already been executed are kept unchanged.

1.4 Objectives

The main objective of this thesis is to develop a resource-aware responder-UAV planning framework for post-earthquake survivor assistance. The framework is designed to use limited responder-UAV resources where they are most useful during the waiting period between survivor detection and ground rescue.

The specific objectives are:

- To jointly determine which survivor-assistance tasks are served, which responder UAVs serve them, in what order they are served, and which feasible 3D routes are used.
- To represent time-critical task priority using operational severity and a time-dependent service value that decreases with service delay.
- To incorporate practical responder limitations including relief-item requirements, parcel and payload capacities, route-dependent payload evolution, battery consumption, reserve energy, and safe return to base.
- To develop a controlled one-for-one task-replacement mechanism for considering newly arrived survivor-assistance tasks during active missions.
- To evaluate the proposed framework through simulation and compare its performance with conventional responder-dispatch strategies.

1.5 Research Contributions

The expected contributions of this thesis are summarized as follows:

- **Proactive Golden-Hour Bridge Model (PGBM):** A planning framework is introduced for using responder UAVs as a temporary assistance bridge between survivor detection and ground rescue.

- **Separation of scout and responder roles:** Scout UAVs are used for upstream reconnaissance, while a dedicated responder fleet is used for relief delivery. PGBM begins after a survivor report has been accepted as an assistance task by the command center.
- **Joint task allocation and 3D routing:** Task selection, UAV assignment, service order, and feasible 3D route selection are considered together, so the routing and resource effects of each assignment are visible during planning.
- **Time-dependent service-value objective:** PGBM combines operational severity with service delay and maximizes the total value obtained from completed assistance tasks.
- **Resource-aware mission planning:** The framework considers relief quantities, parcel and payload capacities, payload evolution, battery-energy consumption, reserve energy, and safe return to base.
- **Controlled dynamic recourse:** A one-for-one task-replacement mechanism allows one newly arrived task to replace one uncompleted task when the replacement is feasible and improves the remaining mission value.

This scope is intentionally limited. PGBM does not propose a new survivor-detection, environmental-mapping, or physical-rescue algorithm. Instead, it operates as a downstream decision layer that uses accepted survivor information and an externally supplied feasible 3D free-space representation.

1.6 Research Challenges

The main challenges addressed in this research include:

- **Joint decision making:** Task assignment and routing are strongly coupled because route length, completion time, payload, and battery usage influence which assignments are feasible.

- **Time-sensitive prioritization:** Higher-priority tasks should generally receive earlier assistance, but the planner must consider the effect of each decision on the total mission value.
- **Three-dimensional route feasibility:** A detected survivor may be located inside rubble or a damaged structure, so the responder must use a nearby feasible drop-off waypoint and travel only through permitted 3D free space.
- **Changing payload and energy use:** Relief payload decreases after deliveries, causing the UAV's carried mass and payload-dependent energy consumption to change throughout the route.
- **Dynamic task arrival:** New survivor-assistance tasks may appear after dispatch, requiring mission adaptation without changing already completed actions.
- **Limited onboard supplies:** An active responder can serve a newly arrived task only when its current onboard inventory is sufficient for the modified remaining mission.

1.7 Organization of the Thesis

The remainder of this thesis is organized as follows:

- **Chapter 2 – Related Work:** Reviews existing research on UAV-assisted disaster response, survivor detection and localization, multi-UAV search, 3D coverage, resource-aware mission planning, and adaptive UAV coordination.
- **Chapter 3 – Proposed Methodologies:** Presents the PGBM system model and mathematical problem formulation, including the time-dependent service-value objective, joint task allocation and 3D routing, payload and battery constraints, and the controlled one-for-one task-replacement mechanism. It also describes the overall operational workflow, responder-mission planning, mission execution, state updates, and dynamic recourse procedure.

-
- **Chapter 4 – Implementation:** Describes the simulation environment, system parameters, optimization implementation, and experimental setup used to evaluate PGBM.
 - **Chapter 5 – Experimental Results:** Evaluates PGBM under different task-arrival, responder, environmental, and resource conditions and compares its performance with selected responder-dispatch strategies.
 - **Chapter 6 – Conclusion:** Summarizes the main findings, discusses the limitations of the proposed framework, and presents possible directions for future research.

Chapter 2

Related Work

UAV-assisted disaster response has been studied from several directions: perception and scene understanding, survivor localization, cooperative search, delivery logistics, resource-aware mission planning, and adaptive UAV coordination. These works provide many of the upstream capabilities needed in a post-earthquake response system. The focus of this thesis is narrower and downstream: once survivor information has been accepted by a command center, PGBM plans how responder UAVs should deliver temporary assistance before ground SAR teams arrive.

2.1 UAV-Based Disaster Perception and Search

One major stream of research studies how UAVs can observe disaster scenes and identify useful response information. Jankovic et al. use an optimized Swin Transformer for UAV-assisted disaster-scene classification [8]. Their work is relevant because it shows how aerial images can support emergency assessment, although the main task is scene classification rather than survivor-assistance planning.

Several works move closer to survivor localization. ZaidAlkilani et al. combine multi-UAV coverage with YOLOv8X-based human detection and geotagged survivor reporting [19]. Che et al. study UAV-assisted 5G-NR localization using

time-of-flight, angle-of-arrival, and received-signal-strength measurements in complex disaster scenarios [4]. Güzey et al. formulate post-disaster victim localization as a cooperative multi-agent problem with energy and communication constraints [7]. Together, these studies show that survivor information can be obtained through visual, radio-based, or cooperative sensing pipelines.

Scene understanding is also important because a detected survivor location may not be directly reachable by a UAV. Sirma et al. introduce D'RespNeT, a post-earthquake UAV dataset and YOLOv8-based instance-segmentation model for detecting collapsed structures, debris, civilians, access openings, and safe UAV landing locations [15]. Such information is useful for choosing accessible drop-off points and feasible operating regions. More generally, low-altitude disaster imagery datasets such as LADI and AIDER support disaster-scene interpretation from aerial images [10, 14].

Another related stream focuses on how scout UAVs should search the affected environment. Aminzadeh and Khoshnood propose a heterogeneous multi-UAV search architecture for post-disaster assessment [3], while Kareem et al. use bio-inspired swarm coordination for thermal survivor search [9]. Leong et al. consider dynamic decentralized 3D urban coverage, where UAVs coordinate over three-dimensional observation viewpoints [11]. These studies are important for the upstream scout-UAV layer assumed in this thesis.

Overall, the perception, localization, and search literature mainly addresses how survivor or environment information can be obtained. PGBM does not replace these methods. Instead, it assumes that the command center has already accepted survivor information and then asks a different question: how should responder UAVs provide temporary assistance after detection but before physical rescue?

2.2 UAV Delivery, Resource-Aware and Adaptive Mission Planning

UAV delivery and mission-planning studies show that task decisions cannot be separated from vehicle limitations. A delivery UAV has finite battery energy, payload capacity, flight time, and route options. These constraints become more important in disaster response because the environment may contain damaged structures, blocked areas, and uncertain access conditions.

Delivery-oriented studies make this point directly. Dorling et al. include payload weight, battery limits, delivery-time constraints, and drone reuse in drone-specific routing models [6]. Cheng et al. model payload-dependent energy consumption in multi-trip drone routing [5]. These works are not written for post-earthquake survivor assistance, but they show why delivery feasibility cannot be separated from carried load and available energy.

Disaster logistics adds another layer to the routing problem. Tureci-Isik et al. study humanitarian aid delivery using parallel truck-drone operations under road-network disruption and uncertain travel times [17]. Liu et al. and Xiong et al. consider UAV mission assignment and path planning for disaster rescue, with Xiong et al. placing particular emphasis on 3D path planning [12, 18]. These studies are useful for PGBM because the responder UAVs must also plan feasible routes in a damaged environment.

Resource-aware emergency-response planning has also been studied beyond physical delivery. Akter et al. consider task offloading, trajectory, and computational-resource allocation in UAV-aided emergency response [1, 2]. Although their tasks are computational rather than relief-delivery tasks, the work supports the broader idea that UAV task decisions and resource limits should be handled together.

Finally, dynamic disaster settings require some ability to react to new information. Sayeed et al. and Güzey et al. study adaptive UAV coordination under uncertain or changing disaster conditions [7, 13]. In those works, adaptation is mainly tied to

tracking or localization. PGBM uses a narrower form of adaptation: a controlled task replacement inside an already planned relief-delivery mission.

2.3 Comparative Analysis and Research Gap

Table 2.1 compares the most closely related studies against the modeling dimensions used in PGBM. The purpose of the table is not to rank the papers, but to show which part of the overall disaster-response pipeline each work mainly addresses.

TABLE 2.1: Comparison of selected UAV disaster-response studies with PGBM.

Study	Main Role	Routing / 3D Aspect	Resource Awareness	Time / Recourse
ZaidAlkilani et al. [19]	Survivor detection and localization	Partial	Partial energy consideration	Not addressed
Sirma et al. [15]	Disaster-scene understanding	Partial 3D scene context	Not a planning model	Not addressed
Dorling et al. [6]	Drone delivery routing	Delivery route planning	Payload and battery considered	Delivery-time limits, no survivor recourse
Xiong et al. [18]	Mission assignment and 3D path planning	Explicit 3D path planning	Partial mission constraints	Not addressed
Akter et al. [2]	Trajectory and computation offloading	Trajectory optimization	Energy and computation resources	Latency-aware, not relief recourse
Güzey et al. [7]	Adaptive victim localization	Map feasibility considered	Energy and communication constraints	Adaptive localization focus
PGBM	Post-detection relief delivery	Feasible 3D responder routes	Payload, energy, reserve, safe return	Time value and task replacement

The comparison suggests three connected gaps. First, there is a post-detection assistance gap. Detection, localization, and scene-understanding studies provide

survivor information, but they usually stop before modeling how temporary assistance should be delivered while ground rescue is delayed.

Second, there is an integrated resource-aware planning gap. Delivery and mission-planning studies consider routing, payload, energy, or disaster logistics, but these elements are usually treated in separate settings. The reviewed work does not jointly address task selection, 3D responder routing, changing payload, battery feasibility, and safe return for post-detection relief delivery.

Third, there is a time-critical and dynamic recourse gap. Existing adaptive UAV studies mainly focus on changing search, tracking, or localization decisions. PGBM instead considers whether a newly arriving survivor-assistance task should replace one uncompleted task in an active relief-delivery mission.

PGBM is therefore positioned as a downstream responder-planning layer. It does not replace scout-UAV detection, survivor localization, environmental mapping, or ground-team physical rescue. It uses accepted survivor-assistance tasks as input and plans how limited responder UAVs should deliver temporary relief during the critical interval between detection and ground rescue.

2.4 Summary

The reviewed studies provide the sensing, localization, coverage, and resource-aware planning foundations required for UAV-assisted disaster response. The remaining need is a compact model for post-detection temporary assistance, where responder UAVs must choose which tasks to serve, how to route through a feasible 3D environment, how to respect payload and battery limits, and how to react when new tasks arrive during an active mission. Chapter 3 develops the proposed PGBM formulation for this downstream planning problem.

Chapter 3

Proposed Methodology

This chapter formulates the responder-UAV planning problem used in the Proactive Golden-Hour Bridge Model (PGBM). The formulation focuses on the delivery layer after survivor reports have already been accepted by the command center. Scout-UAV detection, human or AI-assisted verification, and physical rescue by ground SAR teams are treated as external processes.

The responder planner decides which accepted survivor-assistance tasks should be served in the current planning step, which UAV should serve each selected task, which feasible three-dimensional route should be followed, and whether the resulting missions satisfy payload and battery limits. The formulation also includes a limited recourse rule for newly arriving tasks when all responder UAVs are already committed.

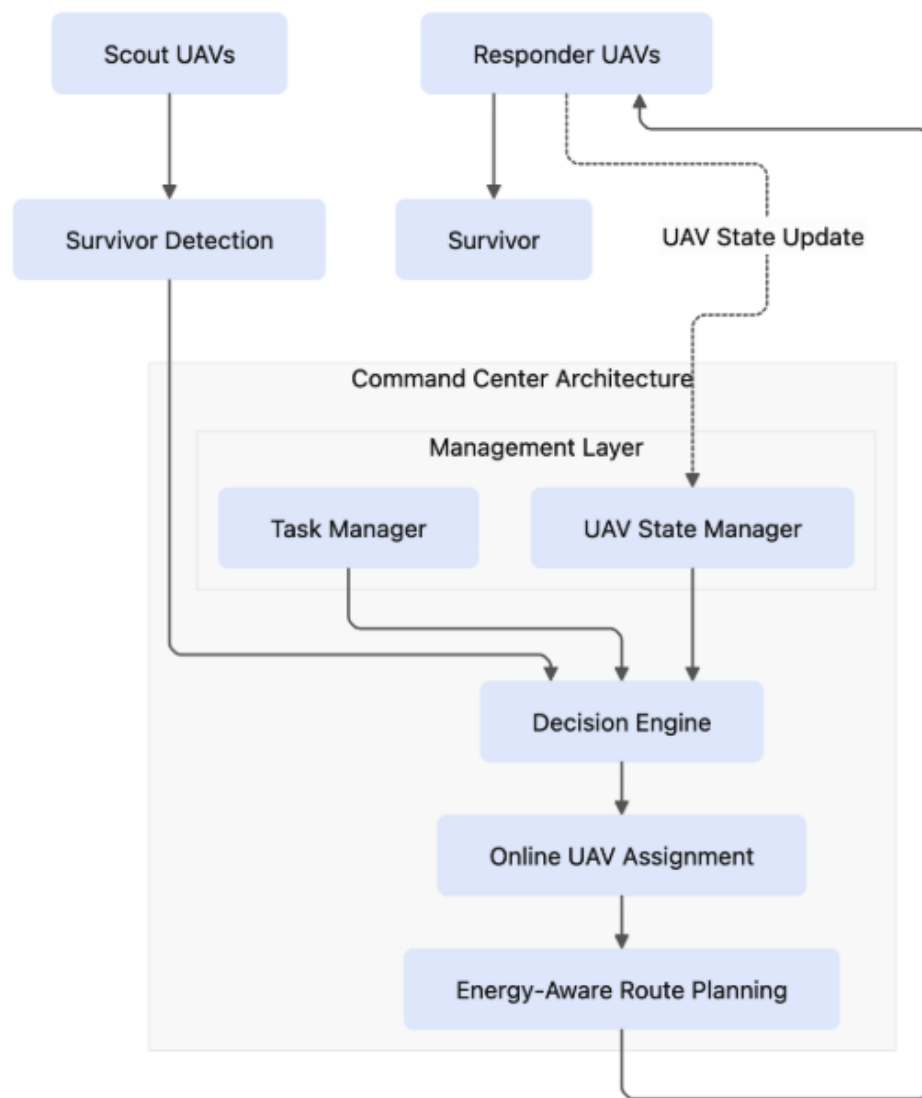


FIGURE 3.1: Overall PGBM system architecture showing scout-UAV detection, command-center task and UAV-state management, online responder assignment, energy-aware route planning, and mission-state feedback.

3.1 Main Notation

Table 3.1 summarizes the main notation used throughout the formulation. Key symbols are also briefly reintroduced near the corresponding equations where needed for clarity.

TABLE 3.1: Main notation used in the PGBM formulation.

Symbol	Meaning
$\mathcal{I}(t)$	Eligible survivor-assistance tasks at planning time t
$\mathcal{U}(t)$	Available responder UAVs at planning time t
$\mathcal{R}$	Relief-item types
$\mathcal{N}$	Feasible UAV waypoints in the flyable 3D operating space
$\mathcal{E}$	Feasible directed flight links between waypoints
i, u, r	Task, UAV, and relief-item indices
m, n	Waypoint indices
x_{iu}	Binary task-assignment decision
y_{mnu}	Binary flight-link selection decision
s_i	Operational severity score of task i
t_i	Detection time of task i
λ	Common temporal decay rate of service value
$v_i(\tau)$	Time-dependent service value of task i at time τ
δ_i	Delivery or service duration at task i
b_u^s, b_u^r	Logical start and return base nodes of UAV u
m_i	UAV-accessible drop-off waypoint for task i
$\mathbf{p}_m$	3D coordinate of waypoint m
d_{mn}	3D distance of flight link (m, n)
t_{mnu}	Travel time of UAV u on link (m, n)
D_u, T_u	Total selected travel distance and travel time of UAV u
τ_{mu}, C_i	Arrival time at waypoint m and completion time of task i
$q_{ir}(t)$	Remaining quantity of item type r required by task i

Symbol	Meaning
w_r	Weight of one item of type r
K_u, W_u	Parcel-count and payload-weight capacities of UAV u
$M_i(t), M_u^0$	Relief mass of task i and initial payload of UAV u
L_{mnu}	Payload mass carried by UAV u on link (m, n)
$E_u(t), E_u^{\text{res}}$	Available and reserve battery energy of UAV u
E_u^{req}	Required mission energy of UAV u
a_{mn}	Positive ascent on flight link (m, n)
$\alpha_u, \beta_u, \gamma_u$	Energy coefficients for UAV u
P_u^{srv}	Service or hovering power of UAV u
$\mathcal{T}_u^{\text{rem}}(t_r)$	Uncompleted tasks of UAV u at recourse time t_r
$J_{ui}^{\text{old}}, J_{ui}^{\text{new}}$	Remaining mission value before and after replacing task i
ΔJ_{ui}	Objective gain from replacing task i
$\mathcal{S}(t_r), \mathcal{Q}(t_r)$	Feasible replacement set and waiting task queue

3.2 Decision Variables

At planning time t , the command center considers a set of eligible tasks $\mathcal{I}(t)$ and a set of available responder UAVs $\mathcal{U}(t)$. The task-assignment decision is

$$x_{iu} = \begin{cases} 1, & \text{if task } i \text{ is assigned to UAV } u, \\ 0, & \text{otherwise.} \end{cases} \quad (3.1)$$

The feasible flight environment is represented by a waypoint set $\mathcal{N}$ and a directed flight-link set $\mathcal{E}$. A waypoint is a candidate UAV position in flyable 3D space, and each waypoint m has coordinate $\mathbf{p}_m = (p_m^x, p_m^y, p_m^z)$. Buildings, rubble, unsafe structures, and no-fly volumes are excluded when these waypoints and flight links are constructed. A flight link (m, n) is included in $\mathcal{E}$ only if the straight segment between the two waypoints remains in flyable space. The routing decision is

$$y_{mnu} = \begin{cases} 1, & \text{if UAV } u \text{ travels from waypoint } m \text{ to } n, \\ 0, & \text{otherwise.} \end{cases} \quad (3.2)$$

The continuous variable τ_{mu} denotes the arrival time of UAV u at waypoint m , while C_i represents the completion time of task i . The variable L_{mnu} represents the payload mass carried by UAV u while traversing flight link (m, n) . These continuous variables satisfy

$$\tau_{mu} \geq 0, \quad C_i \geq 0, \quad L_{mnu} \geq 0. \quad (3.3)$$

3.3 Time-Dependent Service Value

Each accepted survivor-assistance task i is assigned an operational severity score $s_i \in [0, 1]$ by the command center, where a larger value represents higher operational priority. PGBM treats this severity score as an input rather than estimating it. Let t_i denote the detection time of task i , measured from the earthquake occurrence.

The service value of the task at time τ is defined as

$$v_i(\tau) = s_i e^{-\lambda(\tau - t_i)}, \quad \tau \geq t_i, \quad \lambda > 0. \quad (3.4)$$

At the detection time,

$$v_i(t_i) = s_i, \quad (3.5)$$

after which the service value decreases continuously as assistance is delayed.

The parameter λ is the common proportional decay rate; when time is measured in minutes, its unit is min^{-1} . The same proportional decay rate is used for all severity levels, so severity determines the initial task value while delay determines how that value decreases over time. Consequently, at the same delay, a higher-severity task remains at least as valuable as a lower-severity task. Tasks with different delays may nevertheless have different current values, allowing the planner to account for both severity and waiting time. The service value is an operational priority measure and should not be interpreted as a probability of survival. The decay parameter is calibrated experimentally in Section 5.1.

3.4 Optimization Problem

The objective of the responder planner is to maximize the total time-dependent service value obtained from the tasks served in the current planning step:

$$\max \sum_{i \in \mathcal{I}(t)} \sum_{u \in \mathcal{U}(t)} x_{iu} v_i(C_i). \quad (3.6)$$

Using the service-value model defined in Section 3.3,

$$v_i(C_i) = s_i e^{-\lambda(C_i - t_i)}. \quad (3.7)$$

A task contributes to the objective only when it is assigned to a responder UAV. Tasks not selected in the current planning step contribute zero value and remain pending for later consideration. Because service value decreases with completion delay, the objective encourages the planner to serve valuable tasks as early as possible while still respecting routing and resource constraints. The exponential term makes the objective nonlinear, while the assignment and routing decisions are binary. The resulting formulation is therefore a mixed-integer nonlinear optimization problem. The constraints of the model are listed in the following subsections as C1, C2, and so on.

3.5 Joint Task Allocation and Three-Dimensional Routing

For route accounting, the base station of UAV u is written as a logical start node b_u^s and a logical return node b_u^r . Both nodes refer to the same physical base location:

C1: Same-base return condition.

$$\mathbf{p}_{b_u^s} = \mathbf{p}_{b_u^r}, \quad \forall u \in \mathcal{U}(t).$$

For task i , $m_i \in \mathcal{N}$ denotes the UAV-accessible drop-off waypoint. This point may differ from the detected survivor coordinate. For example, if the survivor is inside rubble or a damaged structure, the UAV may not be able to reach the exact survivor position. In that case, m_i is a nearby feasible waypoint from which the relief parcel can be delivered or released safely.

Each task can be assigned to at most one responder UAV:

C2: Single-assignment condition.

$$\sum_{u \in \mathcal{U}(t)} x_{iu} \leq 1, \quad \forall i \in \mathcal{I}(t).$$

If a task is not assigned in the current planning step, it remains pending and can be considered again later.

If UAV u is assigned one or more tasks, its selected route must leave its own start-base node:

C3: Departure-from-base condition.

$$\sum_{i \in \mathcal{I}(t)} x_{iu} \leq |\mathcal{I}(t)| \sum_{n: (b_u^s, n) \in \mathcal{E}} y_{b_u^s, n, u}, \quad \forall u \in \mathcal{U}(t).$$

The same mission must also return to the corresponding return-base node:

C4: Return-to-base condition.

$$\sum_{i \in \mathcal{I}(t)} x_{iu} \leq |\mathcal{I}(t)| \sum_{m: (m, b_u^r) \in \mathcal{E}} y_{m, b_u^r, u}, \quad \forall u \in \mathcal{U}(t).$$

3.6 Travel and Task Completion Time

For each feasible flight link $(m, n) \in \mathcal{E}$, the 3D travel distance is

$$d_{mn} = \sqrt{(p_m^x - p_n^x)^2 + (p_m^y - p_n^y)^2 + (p_m^z - p_n^z)^2}. \quad (3.8)$$

Let ν_u denote the cruise speed of UAV u . The corresponding travel time on link (m, n) is

$$t_{mnu} = \frac{d_{mn}}{\nu_u}. \quad (3.9)$$

The total selected travel distance and travel time for UAV u are

$$D_u = \sum_{(m, n) \in \mathcal{E}} d_{mn} y_{mnu}, \quad (3.10)$$

and

$$T_u = \sum_{(m, n) \in \mathcal{E}} t_{mnu} y_{mnu}. \quad (3.11)$$

These sums include the departure from base, travel between selected drop-off waypoints, and the return-to-base link.

Along a selected flight link, arrival time must increase by at least the corresponding travel time:

C5: Selected-link timing condition.

$$y_{mnu} = 1 \Rightarrow \tau_{nu} \geq \tau_{mu} + t_{mnu}.$$

If task i is assigned to UAV u , its completion time is the arrival time at the drop-off waypoint plus the service duration:

C6: Task-completion time condition.

$$C_i = \tau_{m_i u} + \delta_i, \quad \text{if } x_{iu} = 1.$$

3.7 Relief Capacity and Payload Evolution

Let $\mathcal{R}$ be the set of relief-item types. At time t , task i requires $q_{ir}(t)$ remaining items of type r , and each item of type r has weight w_r . UAV u can carry at most K_u parcels:

C7: Parcel-count capacity condition.

$$\sum_{i \in \mathcal{I}(t)} \sum_{r \in \mathcal{R}} q_{ir}(t) x_{iu} \leq K_u.$$

The same selected load must also satisfy the payload-weight capacity:

C8: Payload-weight capacity condition.

$$\sum_{i \in \mathcal{I}(t)} \sum_{r \in \mathcal{R}} w_r q_{ir}(t) x_{iu} \leq W_u.$$

The relief mass required by task i is

$$M_i(t) = \sum_{r \in \mathcal{R}} w_r q_{ir}(t), \quad (3.12)$$

and the initial payload assigned to UAV u is

$$M_u^0 = \sum_{i \in \mathcal{I}(t)} M_i(t) x_{iu}. \quad (3.13)$$

The payload carried on a flight link can be positive only if that link is selected:

C9: Selected-link payload condition.

$$L_{mnu} \leq W_u y_{mnu}.$$

At departure, the UAV carries the complete relief mass assigned to its mission:

C10: Initial payload condition.

$$\sum_{n: (b_u^s, n) \in \mathcal{E}} L_{b_u^s, n, u} = M_u^0.$$

After task i is served, the carried payload decreases by the mass delivered at that drop-off waypoint:

C11: Payload-reduction condition.

$$\sum_{n: (m_i, n) \in \mathcal{E}} L_{m_i n u} = \sum_{m: (m, m_i) \in \mathcal{E}} L_{m m_i u} - M_i(t) x_{iu}.$$

3.8 Energy and Battery Feasibility

Battery feasibility is checked against the available energy $E_u(t)$ while keeping reserve energy E_u^{res} unused for planned service. For each feasible flight link (m, n) , positive

ascent is

$$a_{mn} = \max(0, p_n^z - p_m^z). \quad (3.14)$$

The required mission energy is modeled as

$$\begin{aligned} E_u^{\text{req}} = & \sum_{(m,n) \in \mathcal{E}} \alpha_u d_{mn} y_{mnu} + \sum_{(m,n) \in \mathcal{E}} \beta_u d_{mn} L_{mnu} \\ & + \sum_{(m,n) \in \mathcal{E}} \gamma_u a_{mn} y_{mnu} + \sum_{i \in \mathcal{I}(t)} P_u^{\text{srv}} \delta_i x_{iu}. \end{aligned} \quad (3.15)$$

The four terms represent baseline travel energy, payload-related travel energy, positive-ascent energy, and service or hovering energy. Since the selected route includes the return-to-base link, the energy required for safe return is included in the same expression.

The mission is feasible only if the required energy is no greater than the usable battery energy:

C12: Energy-feasibility condition.

$$E_u^{\text{req}} \leq E_u(t) - E_u^{\text{res}}.$$

The reserve term leaves a safety margin for operational uncertainty and safe recovery.

3.9 Task-Replacement Recourse

New survivor-assistance tasks may arrive while all responder UAVs are already deployed. In that case, the planner does not restart the full mission plan. Instead, it checks whether the newly arrived task should replace one uncompleted task in an active mission.

Let n denote the new task and let t_r denote the recourse decision time. For UAV u , define

$$\mathcal{T}_u^{\text{rem}}(t_r)$$

as the set of tasks that remain uncompleted at time t_r . Each candidate recourse action replaces one task $i \in \mathcal{T}_u^{\text{rem}}(t_r)$ with the newly arrived task n . The completed part of the mission remains fixed.

Before replacement, the value of the remaining mission is

$$J_{ui}^{\text{old}} = \sum_{j \in \mathcal{T}_u^{\text{rem}}(t_r)} v_j(C_j^{\text{old}}). \quad (3.16)$$

After task i is replaced by task n , the remaining route is recalculated. The new remaining value is

$$J_{ui}^{\text{new}} = \sum_{j \in (\mathcal{T}_u^{\text{rem}}(t_r) \setminus \{i\}) \cup \{n\}} v_j(C_j^{\text{new}}). \quad (3.17)$$

The gain from this replacement is

$$\Delta J_{ui} = J_{ui}^{\text{new}} - J_{ui}^{\text{old}}. \quad (3.18)$$

A replacement is feasible only if UAV u already carries the relief items required by the new task:

$$q_{nr}(t_r) \leq I_{ur}(t_r), \quad \forall r \in \mathcal{R}. \quad (3.19)$$

The modified remaining mission must also satisfy battery feasibility:

$$E_{ui}^{\text{new}} \leq E_u(t_r) - E_u^{\text{res}}, \quad (3.20)$$

and its recalculated route must use feasible flight links in $\mathcal{E}$ and return safely to the UAV base.

Let $\mathcal{S}(t_r)$ be the set of feasible one-for-one replacements at time t_r . The selected replacement is

$$(u^*, i^*) = \arg \max_{(u,i) \in \mathcal{S}(t_r)} \{\Delta J_{ui}\}. \quad (3.21)$$

The replacement is accepted only if it produces a positive gain:

$$\Delta J_{u^*i^*} > 0. \quad (3.22)$$

If accepted, task n is inserted into the remaining mission of UAV u^* , and the displaced task i^* is returned to the waiting queue:

$$\mathcal{Q}(t_r^+) = \mathcal{Q}(t_r) \cup \{i^*\}. \quad (3.23)$$

If no feasible replacement produces a positive gain, the active missions remain unchanged and the newly arrived task is added to the waiting queue:

$$\mathcal{Q}(t_r^+) = \mathcal{Q}(t_r) \cup \{n\}. \quad (3.24)$$

This recourse rule is intentionally limited. It updates one active mission only when doing so improves the remaining service value, and it does not invalidate route segments or deliveries that have already been completed.

Chapter 4

Implementation

4.1 Implementation Overview

The proposed work will be implemented as a simulation-based optimization study rather than as a physical UAV deployment. The implementation will use a Python-based experimental workflow that can be executed locally and later shared through a reproducible notebook environment such as Google Colab or Jupyter Notebook. This choice keeps the experimental setup transparent while allowing the task-allocation, routing, payload, battery, and recourse components of the proposed PGBM formulation to be tested under controlled post-earthquake response scenarios.

The implementation will focus on generating and solving representative responder-UAV planning instances. Each instance will contain an earthquake-affected urban workspace, accepted survivor-assistance tasks, available responder UAVs, relief-item requirements, feasible three-dimensional flight links, and possible newly arriving tasks for recourse evaluation. The purpose of this chapter is therefore to describe the planned data and environment setup that will be used to evaluate the mathematical formulation presented in [Chapter 3](#).

4.2 Data Sources and Scenario Construction

Public disaster-response datasets will be used to support the scenario design, but they will not be treated as complete optimization datasets. DRespNeT is a UAV-based post-earthquake dataset containing high-resolution aerial imagery with detailed annotations of damaged structures, debris, access points, rescue personnel, vehicles, and visible civilians [15]. It provides a realistic reference for understanding post-earthquake urban environments and UAV-assisted search-and-rescue scenarios.

LADI, the Low Altitude Disaster Imagery dataset, contains aerial images collected from real disaster-affected regions and annotated for disaster-related visual features [14]. It will be used as a reference for the visual characteristics and spatial complexity of emergency environments observed from low-altitude aerial platforms. AIDER, the Aerial Image Dataset for Emergency Response, contains aerial images representing emergency conditions such as collapsed buildings, rubble, fires, floods, traffic incidents, and normal scenes [10]. It is useful for supporting disaster-scene classification and emergency-response scenario design.

These public datasets are used primarily to support the design and realism of the disaster scenario. The task, UAV, payload, energy, and dynamic recourse data required by the proposed optimization model will be generated synthetically, because the available public datasets do not jointly provide survivor-assistance demand, responder-UAV battery state, payload state, obstacle-aware three-dimensional routing links, and recourse-event records.

4.3 Synthetic Survivor-Assistance Task Generation

The synthetic task generator will create survivor-assistance tasks that match the inputs required by the proposed formulation. Each task will include a task identifier, detected survivor location, UAV-accessible drop-off waypoint, detection time,

operational severity score, required relief-item quantities, and service duration. The detected survivor location and the drop-off waypoint will be stored separately, since a survivor may be located inside rubble or a damaged structure while the UAV can only deliver from a nearby feasible position in flyable space.

Task arrival times will also be generated so that both initial planning and dynamic recourse can be tested. Tasks known at the beginning of a planning cycle will form the initial task set, while later arrivals will be used to evaluate whether the task-replacement rule can improve the remaining mission value without restarting the entire plan.

4.4 Three-Dimensional Operating Environment

The main experimental environment will be a custom synthetic three-dimensional urban post-earthquake workspace. The workspace will be represented as a bounded 3D region containing UAV base locations, building obstacles, rubble-affected areas, and no-fly volumes. Feasible UAV waypoints will be sampled only from flyable space, and directed flight links will be created only when the straight-line segment between two waypoints does not intersect a blocked region.

This environment is consistent with simulation-based UAV disaster-response studies, where researchers construct 3D disaster rescue scenarios, terrain models, threat sources, and mission points to evaluate assignment and path-planning methods [12, 18]. For the proposed PGBM implementation, the environment will be kept simple enough to support reproducible experiments, while still preserving the key operational constraints: obstacle avoidance, feasible drop-off points, limited battery, limited payload, and return to the same physical UAV base.

4.5 Responder-UAV and Payload Configuration

Each responder UAV will be described by its base location, cruise speed, available battery energy, reserve energy requirement, maximum payload weight, maximum parcel count, and energy-related coefficients. Relief supplies will be represented using a finite set of item types, where each item type has a specified unit weight. The generated task demand will determine how many items of each type must be loaded for a selected task.

The same UAV and payload configuration will be used consistently across the assignment, routing, timing, payload, and energy calculations. In particular, each planned mission must start from the UAV base, visit the assigned drop-off waypoints through feasible flight links, and return to the same physical base while preserving the required reserve energy.

4.6 Optimization and Recourse Workflow

The implementation workflow will begin by generating a disaster scenario and constructing the corresponding 3D flight-link graph. The planner will then create the current set of eligible survivor-assistance tasks and the available responder-UAV states. The optimization model will be solved on this generated instance to obtain the selected tasks, assigned UAVs, selected flight links, completion times, payload usage, and energy-feasible missions.

During simulated mission execution, the state of each UAV and each task will be updated. If a new survivor-assistance task arrives while responder UAVs are already active, the recourse module will evaluate whether replacing one uncompleted task in an active mission improves the remaining service value. Accepted replacements will update the remaining mission plan, while rejected new tasks or displaced old tasks will remain in the waiting queue for a later planning cycle.

4.7 Planned Evaluation Metrics

The planned experiments will evaluate both the initial planning decision and the dynamic recourse behavior. The main output measures will include the total time-dependent service value, number of served tasks, average completion delay, total UAV travel distance, battery-energy usage, payload utilization, number of deferred tasks, number of accepted task replacements, and number of rejected recourse events. Safe-return feasibility will also be checked for each UAV mission.

TABLE 4.1: Planned implementation data components.

Component	Planned Data
Scenario references	DRespNeT, LADI, and AIDER for disaster-scene context
Task instances	Survivor location, drop-off waypoint, severity, demand, and service time
UAV configuration	Base, speed, payload capacity, parcel capacity, battery, and reserve energy
3D environment	Workspace boundary, obstacles, feasible waypoints, and flight links
Recourse events	Newly arriving tasks during active responder-UAV missions
Evaluation outputs	Service value, delay, distance, energy use, payload use, and replacement decisions

Chapter 5

Experimental Results

5.1 Temporal Service-Value Decay Experiment

Before running the full responder-UAV planning experiment, an initial calibration experiment was performed to examine the time-dependent service value function used in the PGBM objective. The purpose of this experiment was to identify a simple severity-aware priority function that decreases with service delay while preserving the ordering between tasks with different operational severity scores.

This experiment was necessary because the service-value function directly controls how the optimization model prioritizes tasks. If only the fixed severity score is used, the model cannot represent the loss of benefit caused by delayed assistance. On the other hand, if the decay function is too aggressive for high-severity tasks, the model may unintentionally allow a lower-severity task to become more valuable than a higher-severity task after some delay. The selected function therefore needs to represent time pressure without violating the intended severity ordering.

The experiment considered seven operational severity levels from 0.3 to 0.9 and evaluated service value over a 120-minute post-detection period. A reference policy table was first prepared to represent a plausible operational assumption: tasks with higher severity start with higher value, and every task loses value as service is delayed. The reference values were not treated as measured medical survival

probabilities; they were used only as a calibration guide for selecting a service-value function.

TABLE 5.1: Reference policy values used for service-value calibration.

Severity	0 min	30 min	60 min	120 min
0.9	0.90	0.65	0.40	0.15
0.8	0.80	0.60	0.38	0.15
0.7	0.70	0.53	0.35	0.14
0.6	0.60	0.46	0.31	0.13
0.5	0.50	0.38	0.27	0.12
0.4	0.40	0.31	0.23	0.11
0.3	0.30	0.25	0.19	0.10

Table 5.1 shows the reduced version of the reference policy used in the experiment. The complete notebook also included intermediate values at 10 and 90 minutes. The table reflects the intended behavior of the service-value model: value decreases with delay, while higher-severity tasks remain higher in value than lower-severity tasks at the same delay.

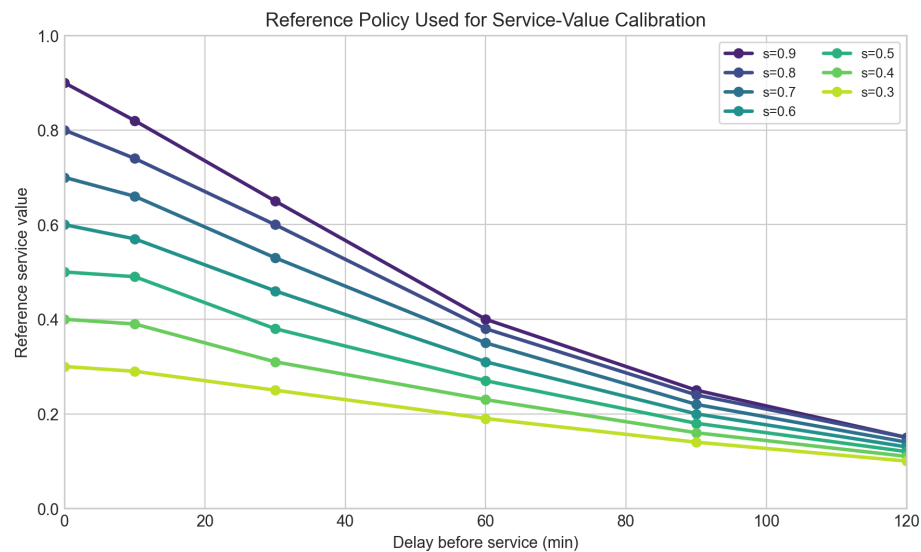


FIGURE 5.1: Reference policy curves used to guide service-value calibration.

Figure 5.1 visualizes the same reference policy values. The graph makes the intended trend clearer than the table alone: each severity level starts from its assigned operational severity score, and the reference value decreases as service delay increases.

Four candidate functions were compared: a common exponential decay function, a severity-dependent exponential decay function, a severity-dependent hyperbolic

decay function, and a severity-dependent linear decay function. For each candidate, the decay parameter was fitted by minimizing the mean squared error between the candidate curve and the reference policy values. After fitting, each function was checked to ensure that service value never increases with delay and that a higher-severity task does not fall below a lower-severity task at the same delay.

TABLE 5.2: Candidate service-value functions evaluated in the experiment.

Candidate	Function Form
Common exponential	$V(s, t) = se^{-\lambda t}$
Severity exponential	$V(s, t) = se^{-\lambda st}$
Severity hyperbolic	$V(s, t) = \frac{s}{1 + \lambda st}$
Severity linear	$V(s, t) = \max(0, s(1 - \lambda st))$

Table 5.2 lists the four candidate forms. Here, s denotes the operational severity score and t denotes the delay after task detection. The common exponential function applies the same decay rate to every task and uses the severity score as the initial value. The severity-dependent functions allow the rate of value loss to change with the severity level.

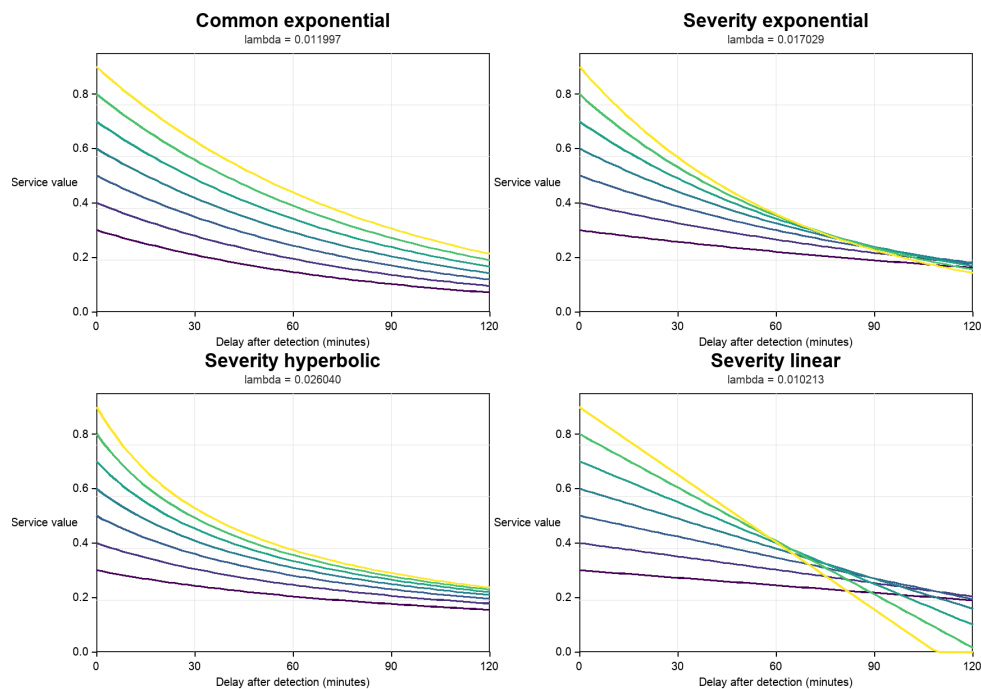


FIGURE 5.2: Fitted service-value curves for the four candidate functions.

Figure 5.2 shows how the four fitted functions behave across the same delay window. The common exponential keeps the curves separated, while the severity-dependent exponential and linear forms allow curves to move closer or cross over time.

TABLE 5.3: Comparison of candidate service-value decay functions.

Candidate Function	Fitted λ	MSE	Ordering
Common exponential	0.011997	0.000892	Preserved
Severity exponential	0.017029	0.001259	Not preserved
Severity hyperbolic	0.026040	0.002742	Preserved
Severity linear	0.010213	0.003129	Not preserved

Table 5.3 shows that the common exponential function produced the smallest fitting error while also satisfying both safety checks. The severity-dependent exponential and linear functions were rejected because they could reverse the intended priority ordering: a lower-severity task could retain more value than a higher-severity task after some delay. The severity-dependent hyperbolic function preserved the ordering, but its fitting error was larger than the common exponential alternative.

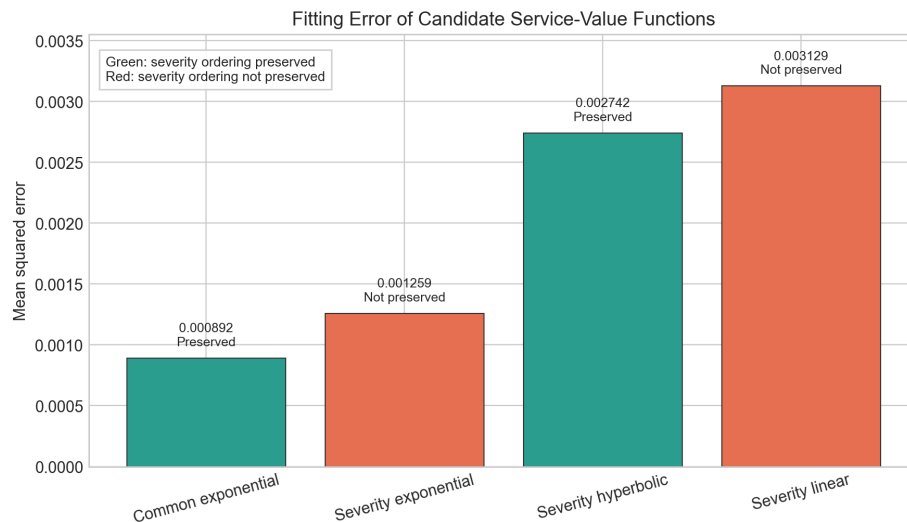


FIGURE 5.3: Fitting error comparison for the candidate service-value functions.

Figure 5.3 summarizes the model selection result. The common exponential candidate has the lowest fitting error and preserves the severity ordering, which makes it the most suitable choice among the tested functions.

The result is important for the proposed PGBM formulation because the selected function must be useful inside an optimization objective, not only visually reasonable in a plot. The common exponential function is smooth, monotone, easy to interpret, and controlled by a single decay parameter. The fitted decay rate also gives a direct way to tune how strongly delay reduces the operational value of a task. For these reasons, it was selected as the most appropriate service-value function for the current model.

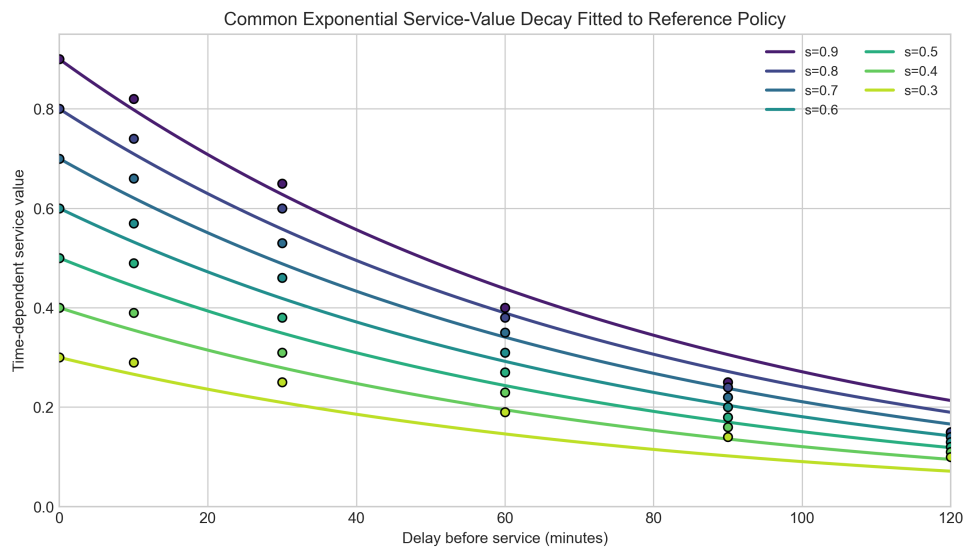


FIGURE 5.4: Fitted common exponential service-value decay curves. Markers indicate the reference policy values used for calibration.

Figure 5.4 illustrates the selected common exponential model. The fitted curves decrease smoothly with delay, and the separation between severity levels is preserved throughout the 120-minute window. For example, with the fitted decay rate, a task with severity 0.9 has service value approximately 0.628 after 30 minutes and 0.438 after 60 minutes, whereas a task with severity 0.6 has value approximately 0.419 after 30 minutes and 0.292 after 60 minutes. Thus, delay reduces the value of all tasks, but the relative priority implied by the operational severity score remains stable.

5.2 Results Analysis

The experiment supports the use of the common exponential service-value model already adopted in the proposed formulation:

$$v_i(C_i) = s_i e^{-\lambda(C_i - t_i)}.$$

This function is simple, monotone, and directly interpretable. The severity score s_i determines the initial operational priority of the task, while the decay parameter λ controls how quickly the value decreases with completion delay. Since the same decay rate is applied to all tasks, the model avoids unintended priority reversals and expresses delay sensitivity through a single tunable parameter.

The selected function should still be interpreted as an operational priority model, not as a validated medical survival model. For the current stage of the thesis, it provides a suitable balance between mathematical simplicity, interpretability, and consistency with the golden-hour motivation.

5.3 Summary of the Experimental Results

The temporal service-value experiment indicates that the common exponential decay function is the preferred service-value model for PGBM. It achieved the lowest mean squared error among the tested candidates and preserved the priority ordering between operational severity levels. Therefore, this thesis uses the common exponential service-value function in the optimization objective. The result justifies the current formulation choice before the full task-allocation, 3D routing, payload, energy, and recourse experiments are implemented.

Chapter 6

Conclusions

6.1 Research Summary

This thesis addresses the delay between UAV-based survivor detection and physical rescue in post-earthquake SAR operations. The proposed PGBM introduces a responder-UAV layer that can deliver lightweight emergency supplies while ground SAR teams are still mobilizing or navigating damaged infrastructure. So far, the work has defined the system scenario, reviewed related UAV disaster response literature, formulated the responder planning problem with task allocation, 3D routing, payload, energy, and recourse constraints, and performed an initial experiment to select a time-dependent service-value function.

6.2 Future Work Plan

Future work will first focus on implementing the proposed framework in a simulation environment, so that the planning behavior can be observed under different disaster-response settings. The model should then be evaluated using clear performance measures and compared with suitable baseline approaches to understand its practical strengths and limitations. Finally, the work can be extended with more realistic data, richer UAV operating constraints, and field-level considerations that would be necessary before any real-world deployment.

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